

Lesson 5: Loops — Do It Again (and Again)!

Hi, coder! 🙌

Imagine clapping 100 times. Would you rather write "clap" one hundred separate times, or just say "clap 100 times" once? That second way is exactly what a **loop** does in coding!

Think Like a Coder

Coders love loops because they let us repeat instructions WITHOUT copying and pasting the same blocks over and over.

- **repeat [10]** — do this a certain number of times, then stop.
- **forever** — keep doing this forever, until the whole program is stopped (click the red stop sign to end it).
- **repeat until** — keep going until something becomes true (we'll dig into this more next lesson!).

Try This: Walk in a Square!

1. Add **"when green flag clicked."**
2. Add a **"repeat [4]"** block.
3. INSIDE the repeat block (in the C-shaped gap), put a **"move 50 steps"** block and a **"turn right 90 degrees"** block.
4. Click the green flag. Your sprite walks in a perfect square — using only ONE move block and ONE turn block, repeated 4 times!

Make It Yours: Loopy Dance

1. Add a **"repeat [8]"** block.
2. Inside it, add a move block, a turn block, and a "next costume" block.
3. Add a **"play sound until done"** block inside the loop too, for a rhythm.
4. Click it and watch your sprite dance in a loop!

★ Bonus Challenge

Try changing the repeat count in your square script — what happens with 3? What about 6? Can you figure out what angle would make a TRIANGLE instead of a square? (Hint: think about 360 divided by the number of sides!)



Debug Detective

Uh-oh! Someone built "repeat 4: move + turn 90" hoping for a square, but their sprite made a weird zig-zag shape instead. What do you think went wrong? Check: were the move and turn blocks actually snapped INSIDE the repeat block's C-shape gap, or did they accidentally land outside it?



Victory Moment

You just learned one of the most powerful ideas in ALL of coding — loops! You built a square-walking sprite and a dancing sprite using way fewer blocks than before. Next time, we'll learn how sprites can make DECISIONS with conditionals. See you in Lesson 6! 🚀